MINI PROJECT REPORT

ON

Student Record Management System

Developed using Java and MySQL

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GITHUB LINK : “https://github.com/esushanthbca25-prog/DBMS_MINI_PROJECT.git”

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I sincerely thank my teachers and project guide for their valuable guidance and support in completing my Student Record Management System using Java and MySQL. I also thank my college, friends, and family for their encouragement and support throughout this project.

ABSTRACT

The **Student Record Management System** is a simple application developed using Java and MySQL. It allows users to register student details such as student ID, name, email, course, and semester. The application stores student information in a MySQL database using JDBC connectivity. This project helps to manage student records efficiently and provides practical knowledge of Java programming and database management.

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1. INTRODUCTION

The Student Record Management System is a simple application developed using Java and MySQL. It helps users store and manage student details such as student ID, name, email, course, and semester.Java is used to develop the application, while MySQL stores the student information in a database. JDBC is used to connect Java with MySQL.This project makes student record management easier, faster, and more organized.

2. OBJECTIVES

The main objectives of the Student Record Management System are:

1. To store student details in a MySQL database.
2. To register new students easily.
3. To view and manage student records.
4. To connect Java with MySQL using JDBC.
5. To reduce manual work and improve record management.

3. SCOPE OF THE PROJECT

The project helps users manage student information through a Java application connected to a MySQL database. It supports registering students, viewing records, updating details, and deleting records. It is useful for learning basic database management and Java programming.

4. SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements	Hardware Requirements
• Operating System: Windows • Programming Language: Java • IDE: Apache NetBeans • Database: MySQL • Database Tool: MySQL Workbench • Connectivity: JDBC MySQL Connector	• Processor: Intel or equivalent • RAM: Minimum 4 GB • Storage: Minimum 500 MB free space • Keyboard and mouse

5.SYSTEM DESIGN AND DATABASE STRUCTURE

The application uses Java for the user interface and MySQL for storing student records. JDBC connects the Java application to the database.

Database Name: student_db

Table Name: students

Column Name	Data Type	Description
student_id	INT	Student ID and primary key
name	VARCHAR(50)	Student name
email	VARCHAR(100)	Student email
course	VARCHAR(50)	Course name
semester	INT	Semester number

6. IMPLEMENTATION

The application is developed using Java in Apache NetBeans and connected to MySQL through JDBC.

The main operations are:

1. **Insert:** Register a new student.
2. **Select:** View student records.
3. **Update:** Modify existing student details.
4. **Delete:** Remove a student record.

The application uses SQL queries and PreparedStatement to perform database operations.

7.TESTING AND RESULTS

The application is tested to verify that student records are stored and managed correctly.

Test Case	Expected Result
Enter valid student details	Student record is saved
View student records	Records are displayed
Update student details	Record is modified
Delete a student record	Record is removed
Enter an existing student ID	Duplicate ID is rejected

The application can successfully manage student records when the database connection and input details are correct.

8. SCREENSHOTS

The following screenshots provide evidence of the application and its MySQL database operations.

8.1 Registration Form

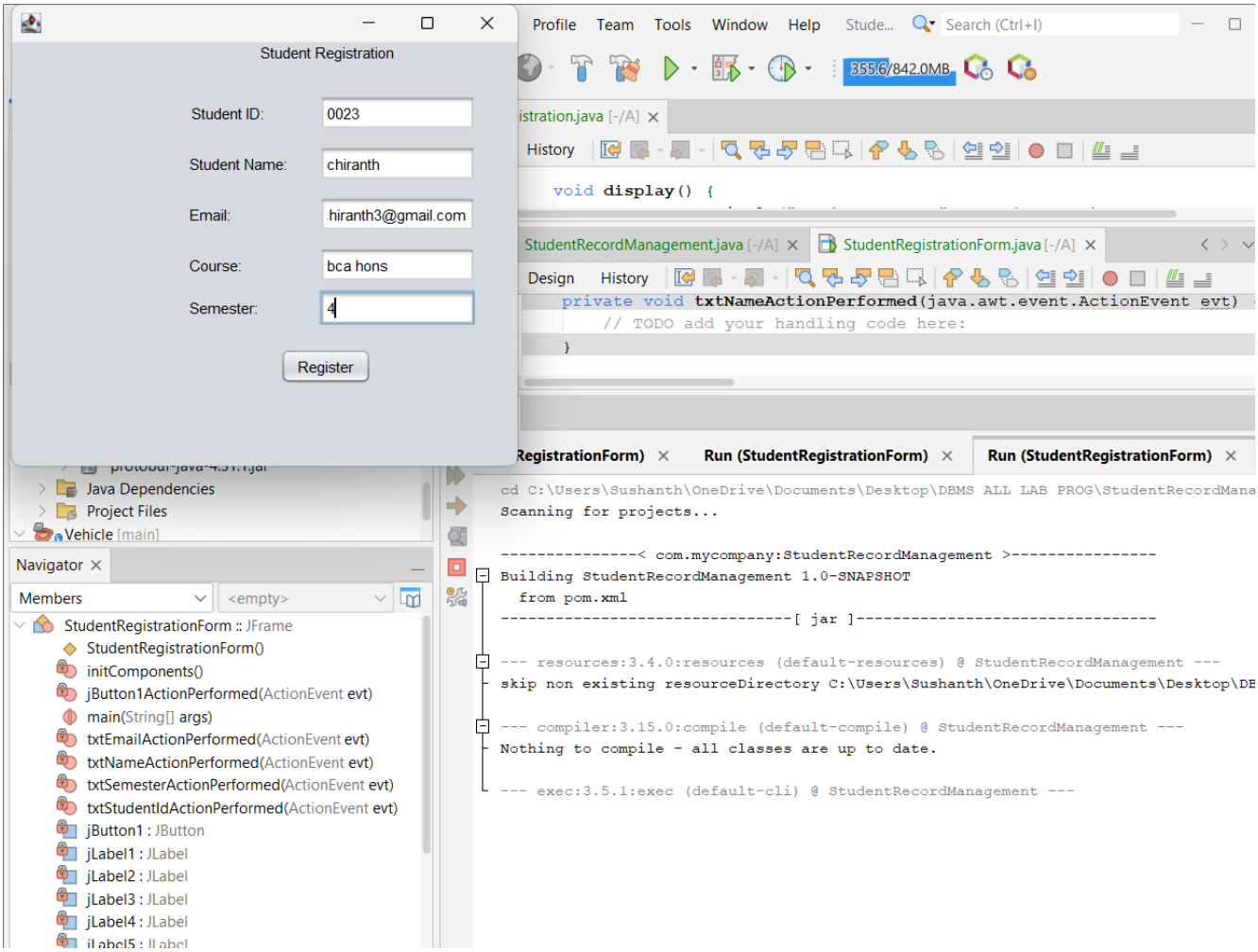


Figure 1: Student Registration Form of the Student Record Management System.

8.2 Successful Registration

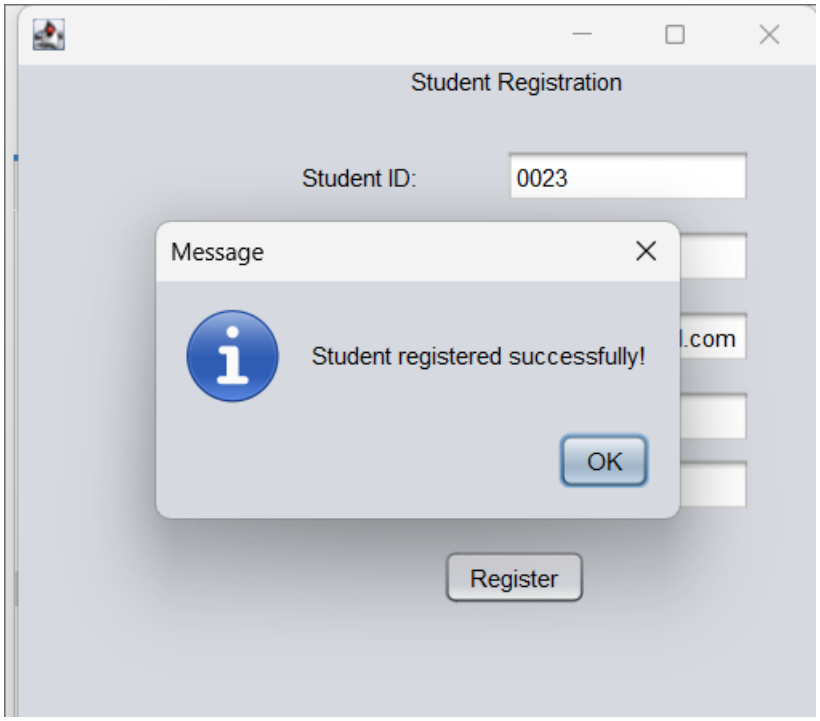


Figure 2: Confirmation message shown after successful registration.

8.3 MySQL Registration Records

Output | Result #2

student_id	name	email	course	semester
23	chiranth	chiranth2@gamil.com	bca	4
30	E Sushanth	esushanth018@gmail.com	BCA Hons	3
101	deepak	deepak@gmail.com	BCA Hons	3
102	sushanth	sushanth2@gmail.com	bca hons	5

Figure 3: Registration records retrieved from MySQL, including the newly registered participant.

8.4 Database Setup

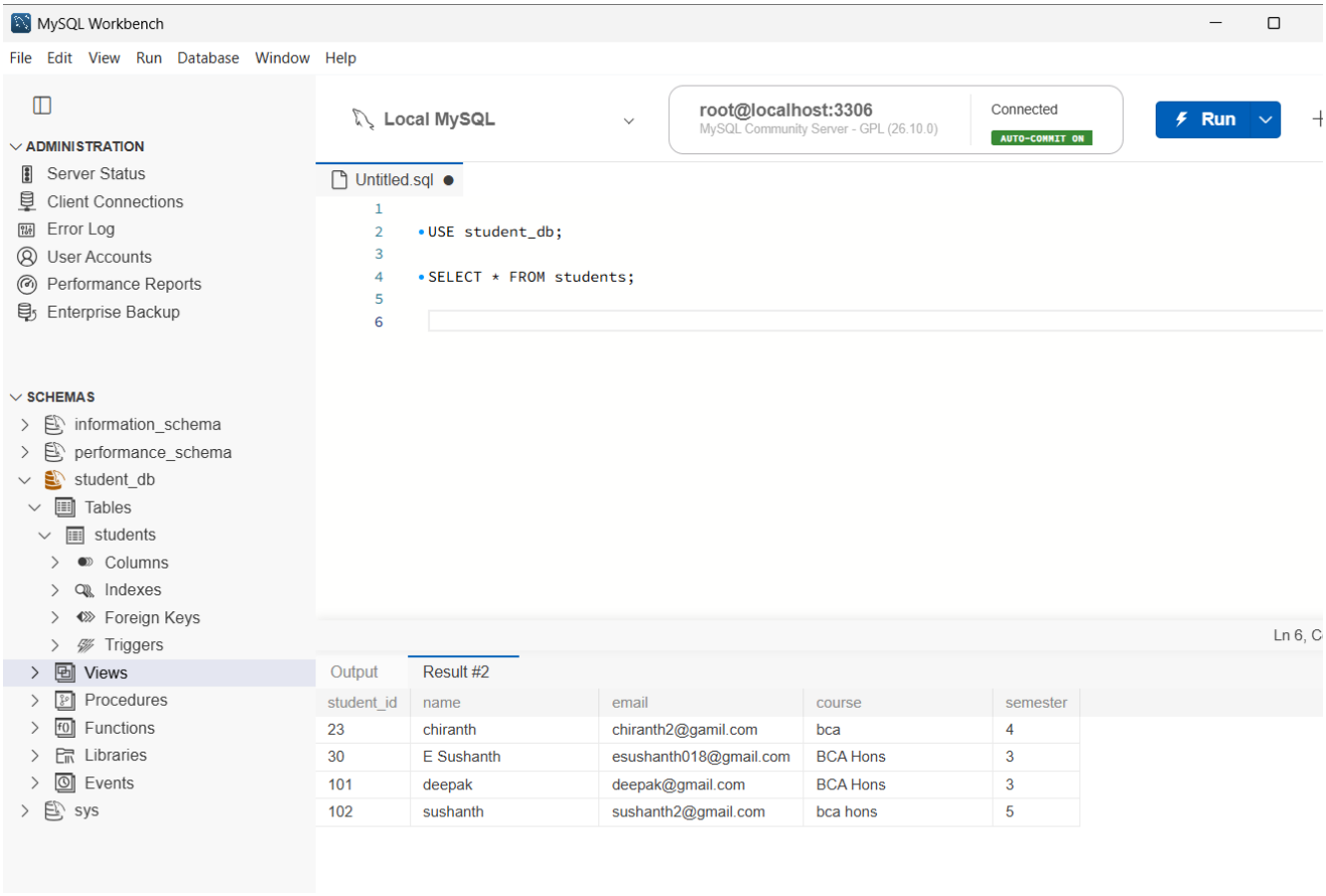


Figure 4:: Student Records Displayed in MySQL Workbench..

9. LIMITATIONS AND FUTURE ENHANCEMENTS

Limitations

- The application requires a working MySQL connection.
- It provides only basic student record management.
- User authentication may not be available.

Future Enhancements

- Add login and password authentication.
- Add student search functionality.
- Improve the user interface.
- Generate student reports.
- Add input validation and better error handling.

10. CONCLUSION

The Student Record Management System was developed using Java and MySQL to manage student information. The project demonstrates database connectivity using JDBC and basic SQL operations such as INSERT, SELECT, UPDATE, and DELETE. It also provides practical experience in Java application development and database management.